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Nambur, Pedakakani (M), Guntur (Dt) - 522508
Department of CSE - Artificial Intelligence & Machine Learning

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category	Research oriented	**Batch ID**	CSM - C13

Title of the Project: Securing Autonomous AI Agents Against Indirect Prompt Injection Attacks

ABSTRACT

Securing autonomous AI agents against malicious instructions remains a significant challenge due the increasing use of Large Language Models (LLMs) in applications that interact with external and potentially untrusted sources such as websites, emails, documents, and APIs. Unlike traditional software systems, autonomous AI agents can interpret natural language, process external information, and dynamically perform actions using connected tools. However, when third-party content contains hidden adversarial instructions, these agents may fail to distinguish between legitimate data and malicious commands. This vulnerability, known as **Indirect Prompt Injection (IPI),** can cause an agent to deviate from the user's original objective, leak sensitive information, or execute unauthorized actions. Existing research has primarily demonstrated the feasibility and impact of indirect prompt injection attacks, highlighting the need for robust architectural approaches that provide effective mitigation while preserving the useful capabilities of autonomous AI agents.

To address these limitations, this project proposes a **Dual-Agent Defense System for Securing Autonomous AI Agents Against Indirect Prompt Injection Attacks**. The proposed framework separates data processing from action execution through three major components: a **Perception Agent, Security Verifier, and Executive Agent**. The Perception Agent processes third-party content in an isolated, read-only environment with no execution authority or direct access to write APIs. It analyzes external content and generates a structured response or proposed tool-call action. The proposed action is then forwarded to the Security Verifier, which acts as a security control layer between untrusted information and privileged agent actions.

The Security Verifier employs a **Dual-Gate Security Verification mechanism** to determine whether a proposed action is safe for execution. The first gate performs **Semantic Intent Drift Analysis**, where the original user goal and the proposed action are represented using embedding vectors. Their semantic similarity is calculated using **Cosine Similarity**, and actions whose similarity falls below a predefined threshold are identified as potential deviations from the user's intended objective. The second gate performs **Deterministic Policy Enforcement** using **Open Policy Agent (OPA)** and declarative Rego rules. These policies are designed to prevent security-sensitive operations such as sensitive-token transfer, unauthorized domain routing, and restricted file-system modifications.

Based on the verification results, the system dynamically routes the proposed action either to a **Security Alert Node** or to the execution stage. If the action is identified as malicious or violates a security policy, execution is immediately terminated, the incident is logged, and a security warning is presented to the user. If the action successfully passes both security gates, it is validated and converted into a strict JSON action schema before being forwarded to the **Executive Agent.** The Executive Agent then performs the authorized API or tool call and returns the safe result to the user.

The proposed framework aims to provide an effective and explainable architectural defense against indirect prompt injection by combining **LLMs, embedding-based semantic analysis, agent isolation, multi-agent orchestration, and deterministic policy enforcement**. The system can be evaluated using publicly available prompt-injection datasets, including the **Open Prompt Injection Dataset** and **NVIDIA Nemotron-RL-Agentic-Indirect-Prompt-Injection-v1**, to assess its ability to identify malicious actions while allowing legitimate agent operations. The project ultimately aims to improve the security and reliability of autonomous AI agents by ensuring that external untrusted content cannot directly influence privileged actions without passing through explicit semantic and policy-based verification.

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References (Category specific):

1. https://arxiv.org/pdf/2302.12173
2. https://arxiv.org/pdf/2605.03378
3. https://arxiv.org/pdf/2509.14285
4. https://arxiv.org/pdf/2406.13352
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Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A42G4	SHAIK MOMINA	
2	23BQ1A42H0	TALLURI YOGESH CHANDRA	
3	23BQ1A42H4	THOKALA VENKATA SIVA GOPI	
4	23BQ1A42E2	PODILA SASWITHA RANI	

Project Guide

Name	Mrs. Shaik Rubeena	Signature	